isolates the mode at the top of the column *Quantity*.

If the contents of the *orderitem* table are as follows:

ID	Order_ ID	Prod- uct_ID	Quantity
1	1	1	3
2	2	5	8
3	3	4	5
4	4	6	7
5	5	4	3
6	6	2	9

Then the mode-query will return the following:

quantity	count
3	2

Measures of variability

In statistics, variability is the extent to which a distribution is extended or squeezed. Common examples of measures of statistical variability are the max., min., variance, standard deviation, and interquartile ranges.

Max. and min.: The maximum and minimum values of any numerical field of a table can be reported in a tabular form by the aggregation functions *max* and *min* of MySQL. To find the *max* and *min* freight values of the invoice table of the Northwind database, the SQL query can be written as follows:

```
Variability
SELECT
MAX(freight) as MAX_
freight,MIN(freight) as MIN_freight
FROMinvoices;
```

Standard deviation: Standard deviation is a measure of the spread of values in a data set. It shows the extent of variation of data from the average (mean). A low standard deviation shows that the values in the data set are close to the mean, while a high standard deviation indicates that the values of the data set are spread out over a large range of values.

MySQL supports both the population and sample standard deviations. There are a few variations among different functions, regarding which readers may consult documents.

STD(field) returns the population standard deviation of the expression and returns NULL if there is no matching row:

```
use northwind;
select format(std(freight),2) as Pop_STD
from orders  as t
where customerID='VINET'
order by customerID;

Output of population standard deviation:
    10.84
```

STDDEV_SAMP(field) returns the sample standard deviation:

```
use northwind;
select format(STDDEV_SAMP(freight),2) as
Pop_STD
from orders  as t
where customerID='VINET'
order by customerID;


Output sample standard deviation:
    12.11
```

The population and sample standard deviations of freight charges of the customer Vinet are shown in the above two code snippets.

MySQL also has functions for population variance and sample variance calculations. These are:

- VAR_POP(field), which calculates the population standard variance of the field.
- VARIANCE(field), which is the same as VAR_POP function.
- VAR_SAMP(field), which calculates the sample standard variance of the field.

Distribution pattern

Distribution of data plays an important role in statistical data analysis. The histogram is an important statistical tool to understand the distribution pattern. The count of items in each group can be done by *count()* of the target item using the *GROUP_BY* clause. This is shown in the following query:

```
SELECT
a.productID,b.ProductName,count(a.
productID) as Product_Count
FROM invoices as a
Join products as b on a.ProductID=b.
ProductID
Group by a.productID
Order bya.productID;
```

The Frequency table is given below.

Pro- ductID	Product- Name	Product_ Count
1	Chai	38
2	Chang	44
3	Aniseed Syrup	12
4	Chef Anton's Cajun Sea- soning	20
5	Chef Anton's Gumbo Mix	10
6	Grandma's Boysenberry Spread	12
7	Uncle Bob's Organic Dried Pears	29
8	Northwoods Cranberry Sauce	13
9	Mishi Kobe Niku	5
10	Ikura	33
11	Queso- Cabrales	38

END

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